

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 4+ years building distributed systems and cloud infrastructure for SaaS products. Led zero-downtime migrations of 20+ production applications from Parse/MongoDB to Firebase/GCP, implementing real-time data pipelines with Pub/Sub and deploying to AWS/GCP at scale. Strong background in backend architecture, microservices design with Apache Kafka and Kubernetes, and cross-functional collaboration with product teams. Proven track record delivering production deployments that drive measurable business impact.

CORE COMPETENCIES

Distributed Systems	AWS / GCP Cloud Infrastructure	Zero-Downtime Migrations	Real-Time Data Pipelines
Microservices Architecture	Production Deployments	SaaS Product Development	Backend Engineering

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer · US Remote

- Implemented real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub for distributed message processing, enabling parallel production traffic during migration
- Led end-to-end migration of 20+ applications from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to CTO
- Saved \$5,000/month in infrastructure costs by migrating all services from AWS to GCP
- Built 3D visualization feature using video overlay and ffmpeg, increasing buyer conversion
- Implemented Express Shipping feature generating additional \$40 revenue per order

MTailor Mar 2022 – Jul 2022

Contractor · US Remote

- Built migration scripts to read data from MongoDB and write to Firestore after complex processing logic
- Wrote documentation for engineers on working with Firebase SDKs (Node.js and Python)
- Trained Ops team on Firebase Dashboard and backup procedures

Dr Wealth Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Built batch jobs querying 2M+ Firestore records to keep customer prices current from Morningstar APIs
- Extended backend APIs using Firebase Cloud Functions and Express.js
- Added responsive UI pages to PWA delivering real-time stock market prices

Mind2matter.co Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications using Web3 and Node.js
- Delivered React UIs from Figma designs under tight timelines, reporting to CTO

CodeBits Jan 2020 – Jul 2021

Software Engineer · Uganda Remote

- Architected microservices backend for FIDA Uganda's case management app using Apache Kafka, Docker, and Kubernetes
- Led team of 4 developers building legal tech systems for NGOs in Uganda
- Built USSD service communicating with Node.js backend via gRPC for legal aid providers
- Developed cross-platform mobile app using React Native and Expo with Firebase push notifications

PROJECTS

Parse → Firebase Migration (MTailor)

Migrated 20+ production applications with zero downtime. Built real-time MongoDB ↔ Firestore sync pipeline supporting parallel traffic during migration. Saved \$5K/month infrastructure costs.

Node.js, Google Pub/Sub, Firebase, GCP, AWS

Real-Time Stock Price PWA (Dr Wealth)

Optimized PWA querying 2M+ Firestore records for real-time stock market data from Morningstar APIs.

Firebase, Node.js, React, Express.js

Legal Case Management System (FIDA Uganda)

Microservices architecture for NGO case management with event-driven design.

Apache Kafka, Docker, Kubernetes, Node.js, gRPC

EDUCATION

Bachelor of Science in Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

TECHNICAL SKILLS

Languages: JavaScript/Node.js, Python, Go, SQL **Cloud:** AWS (S3, EC2, EBS), GCP (Pub/Sub, Cloud Functions), Firebase

Databases: MongoDB, Firestore, SQL **Infrastructure:** Docker, Kubernetes, Apache Kafka, gRPC

Frontend: React, React Native, HTML/CSS, Tailwind